Is our ns-3 setup "O-RAN compliant"?

ns-3 transfer experiment · 31 October 2026

Short answer: no. The radio part is realistic and mostly follows the 5G standards (3GPP). The O-RAN part is only *imitated*: the "xApps" and the "arbiter" are small pieces of code inside the simulator, not real O-RAN software talking over real O-RAN interfaces.

1. What is realistic

The simulator (ns-3 with the 5G-LENA add-on) models a real 5G radio network: how signals are sent and scheduled, how users share the radio, and how cells interfere with each other. Signal loss over distance uses the standard 3GPP city model at 3.5 GHz, with cells 300 m apart.

2. Where we simplified things

With the full-detail settings, one run would take about 20 hours. To bring that down to about 2 hours, we simplified:

What	What we did (and the usual standard choice)
Signal fading	A simpler fading model; the full 3GPP model is much slower.
Antennas	One all-direction antenna per site, instead of the usual three directional sectors.
Bandwidth	5 MHz. We believe real 3.5 GHz networks use at least 10 MHz; to be checked against TS 38.101-1.
Power and users	43 dBm and 3 users per cell; typical studies use more of both.
Handover	Users never switch cells; the simulator does not support it.

3. What is not real O-RAN

Real O-RAN	Our setup
A RIC platform runs the xApps.	No RIC. The xApps are functions inside the simulator.
xApps talk to the network over the E2 interface (KPM to read, RC to control).	No E2. The code changes the transmit power directly.
Measurements arrive as standard E2 reports.	The simulator computes them itself; energy comes from a formula.
xApps act every 10 ms to 1 s ("near-real-time").	Decisions every 20 s, which in O-RAN terms is the slower rApp range. The paper's own simulator uses 60 s.
Conflicts between apps are handled by the RIC's procedures.	A simple fixed priority (Coverage beats Power-saving), same idea, not the official procedure.

4. Does it matter for this experiment?

Mostly no. The question we are testing is: *when the arbiter's choices are partly random and logged, can PMRT find the real effects and stay quiet when there are none?* That depends on the network

reacting realistically to power changes. It does not depend on how the messages are delivered. ns-3 is actually *more* realistic on the radio side than the paper's own simulator.

But be careful how we describe it. We should not call it "O-RAN-compliant". A fair description is: *"a 5G system-level simulation (ns-3 / 5G-LENA) with an emulated RIC control loop"*, with the simplifications above listed as limitations.

5. If we want real O-RAN later

The usual tool is **ns-O-RAN**. It connects ns-3 to a real RIC (from the O-RAN Software Community or OpenRAN Gym) over E2, so the xApps become real xApps. It needs much more setup and each run would be slower. It is best treated as a follow-up study, not a switch in the middle of the current runs, which are about 10 hours from done.